

RAG Grounding Test Cases

Categories 1 to 5

RAG Answer Grounding / Hallucination Detection Evaluation

Prepared from the provided Streamlit pipeline screenshots

Reference Context: Renewable Energy - Solar and Wind Power

1. Purpose and Scope

This document consolidates the RAG answer grounding test-case evidence from Category 1 to Category 5. It is structured as an official validation document where each test case includes the question, expected output, actual output, result, grounded ratio, RAGAS score, and screenshot evidence inside the same table.

2. Test Coverage Overview

Category	Type	Description	Test Cases
Category 1	Fully Grounded	Questions answerable directly from the uploaded context.	15
Category 2	Partially Grounded	Multi-part questions with both supported and unsupported parts.	16
Category 3	Out of Context	Questions unrelated to the document or asking facts not provided.	14
Category 4	Mixed Multi-Part Stress Test	Questions combining valid and invalid parts to test hallucination removal.	5
Category 5	Edge Cases	Trick, ambiguous, and related-sounding prompts.	9
Total	All Categories	Complete screenshot evidence set.	59

3. Result Field Definitions

Field	Meaning
Expected Output	The correct behavior expected from the pipeline for the given question.
Actual Output	The observed behavior shown in the Streamlit test screenshot.
Grounded Claims	Claims supported by the retrieved document context.
Hallucinated Claims	Claims not supported by the document and therefore removed or blocked.
Grounded Ratio	Percentage of grounded claims in the generated answer.
RAGAS Faithfulness	Additional faithfulness score displayed in the local Streamlit interface.

Category 1 - Fully Grounded Test Cases

Single-part questions supported by the uploaded renewable-energy context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-01	What is solar energy?	Answer should state that solar energy is generated by capturing sunlight using photovoltaic (PV) panels that convert sunlight directly into electricity.	The system returned the same PV-panel based explanation and marked it fully grounded.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-01

localhost:8501

Deploy

Run Pipeline

 Results

✓ FULLY GROUNDED

Grounded ratio: 100% · Answer modified: No — fully grounded · RAGAS PASS

1

GROUNDING CLAIMS

0

HALLUCINATED CLAIMS

100%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

GROUNDING CLAIMS

FINAL ANSWER (GUARANTEED GROUNDED)

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

- Solar energy is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity.

Part 1: Solar energy "is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity."

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-02	What are the disadvantages of wind turbines?	Answer should mention only supported disadvantages: wind turbines can be noisy and may affect local wildlife, especially birds and bats.	The system returned the supported disadvantage statement only.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-02

localhost:8501

Deploy

>>

What are the disadvantages of wind turbines?

Run Pipeline

 Results

✓ FULLY GROUNDED

Grounded ratio: 100% · Answer modified: No — fully grounded · RAGAS PASS

1

GROUNDING CLAIMS

0

HALLUCINATED CLAIMS

100%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore. Wind energy is efficient and produces no direct

GROUNDING CLAIMS

- The disadvantages of wind turbines are that they can be noisy and may affect local wildlife, especially birds and bats.

FINAL ANSWER (GUARANTEED GROUNDED)

Part 1: The disadvantages of wind turbines are that they "can be noisy and may affect local wildlife, especially birds and bats."

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-03	How do wind turbines affect wildlife?	Answer should state that wind turbines may affect local wildlife, especially birds and bats.	The system returned only the wildlife impact statement from the context.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-03

localhost:8501

Deploy

How do wind turbines affect wildlife?

Run Pipeline

Results

✓ FULLY GROUNDED

Grounded ratio: 100% · Answer modified: No — fully grounded · RAGAS PASS

1

GROUNDING CLAIMS

0

HALLUCINATED CLAIMS

100%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore.

GROUNDING CLAIMS

- Wind turbines can be noisy and may affect local wildlife, especially birds and bats.

FINAL ANSWER (GUARANTEED GROUNDED)

Part 1: Wind turbines can be noisy and may affect local wildlife, especially birds and bats.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-04	What role do batteries play in renewable energy?	Answer should state that batteries store excess energy and provide power when production is low.	The system returned the supported battery-storage role.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-04

localhost:8501

Deploy

What role do batteries play in renewable energy?

Run Pipeline

Results

✓ FULLY GROUNDED

Grounded ratio: 100% · Answer modified: No — fully grounded · RAGAS PASS

1

GROUNDING CLAIMS

0

HALLUCINATED CLAIMS

100%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Both solar and wind energy play a significant role in reducing reliance on fossil fuels. Governments around the world are investing heavily in renewable energy technologies to combat climate change and reduce carbon emissions. Energy storage systems, such as batteries, are often used alongside renewable energy sources to store excess energy and provide power when production is low.

✓ GROUNDED CLAIMS

- Batteries play a role in renewable energy as energy storage systems that are often used alongside renewable energy sources to store excess energy and provide power when production is low.

FINAL ANSWER (GUARANTEED GROUNDED)

Part 1: Batteries play a role in renewable energy as "energy storage systems" that are "often used alongside renewable energy sources to store excess energy and provide power when production is low."

91°F
Mostly sunny

Search

12:06 P
4/29/202

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-05	What is the role of energy storage systems?	Answer should state that energy storage systems store excess energy and provide power when production is low alongside renewable energy sources.	The system returned the supported storage-system role.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-05

localhost:8501

Deploy

What is the role of energy storage systems?

Run Pipeline

Results

✓ FULLY GROUNDED

Grounded ratio: 100% · Answer modified: No — fully grounded · RAGAS PASS

1

GROUNDED CLAIMS

0

HALLUCINATED CLAIMS

100%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Both solar and wind energy play a significant role in reducing reliance on fossil fuels. Governments around the world are investing heavily in renewable energy technologies to combat climate change and reduce carbon emissions. Energy storage systems, such as batteries, are often used alongside renewable energy sources to store

✓ GROUNDED CLAIMS

The role of energy storage systems is to store excess energy and provide power when production is low alongside renewable energy sources.

FINAL ANSWER (GUARANTEED GROUNDED)

Part 1: The role of energy storage systems is to "store excess energy and provide power when production is low" alongside renewable energy sources.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-06	Why are governments investing in renewable energy?	Answer should state that governments are investing to combat climate change and reduce carbon emissions.	The system returned the same supported reason.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-06

localhost:8501

Deploy

Why are governments investing in renewable energy?

Run Pipeline

 Results

✓ FULLY GROUNDED

Grounded ratio: 100% · Answer modified: No — fully grounded · RAGAS PASS

1

GROUNDED CLAIMS

0

HALLUCINATED CLAIMS

100%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Both solar and wind energy play a significant role in reducing reliance on fossil fuels. Governments around the world are investing heavily in renewable energy technologies to combat climate change and reduce carbon emissions.
Energy storage systems, such as batteries, are often used alongside renewable energy sources to store

GROUNDED CLAIMS

- Governments around the world are investing heavily in renewable energy technologies to combat climate change and reduce carbon emissions.

FINAL ANSWER (GUARANTEED GROUNDED)

Part 1: Governments around the world are investing heavily in renewable energy technologies to combat climate change and reduce carbon emissions.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-07	What makes sunlight a sustainable energy source?	Answer should state that sunlight is abundant and renewable.	The system returned that sunlight is sustainable because it is abundant and renewable.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-07

localhost:8501

Deploy

>>

What makes sunlight a sustainable energy source?

Run Pipeline

Results

✓ FULLY GROUNDED

Grounded ratio: 100% · Answer modified: No — fully grounded · RAGAS PASS

1

GROUNDED CLAIMS

0

HALLUCINATED CLAIMS

100%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation. It is also a sustainable source of energy because sunlight is abundant and renewable. However, solar energy generation depends on weather conditions and is less effective during cloudy days or at night.

✓ GROUNDED CLAIMS

- Sunlight is a sustainable source of energy because it is abundant and renewable.

FINAL ANSWER (GUARANTEED GROUNDED)

Part 1: Sunlight is a sustainable source of energy because it is "abundant and renewable".

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-08	How is solar energy generated?	Answer should state that solar energy is generated by capturing sunlight using PV panels, which convert sunlight directly into electricity.	The system returned the supported generation explanation.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-08

Configuration

Groq API Key

.....

LLM Model

llama-3.3-70b-versatile

Retrieval Settings

Chunk size (chars)

500

Chunk overlap (chars)

50

Top-k chunks to retrieve

5

How is solar energy generated?

Run Pipeline

Results

FULLY GROUNDED

Grounded ratio: 100% · Answer modified: No — fully grounded · RAGAS PASS

2

GROUNDED CLAIMS

0

HALLUCINATED CLAIMS

100%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly

GROUNDED CLAIMS

- It is generated by capturing sunlight using photovoltaic (PV) panels.
- Photovoltaic (PV) panels convert sunlight directly into electricity.

FINAL ANSWER (GUARANTEED GROUNDED)

It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-09	Where are solar panels commonly installed?	Answer should state that solar panels are commonly installed on rooftops or in large solar farms.	The system returned the supported installation locations.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-09



>> Deploy ⋮

? Step 2 — Ask a Question

Where are solar panels commonly installed?

 Run Pipeline

Results

✓ FULLY GROUNDED

Grounded ratio: 100% · Answer modified: No — fully grounded · RAGAS PASS

1

GROUNDING CLAIMS

0

HALLUCINATED CLAIMS

100%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

 RETRIEVED CONTEXT

Title: Renewable Energy — Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic

✓ GROUNDED CLAIMS

- Solar panels are commonly installed on rooftops or in large solar farms.

 FINAL ANSWER (GUARANTEED GROUNDED)

Part 1: Solar panels are commonly installed on rooftops or in large solar farms.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-10	What are the advantages of solar energy?	Answer should mention only supported advantages: clean operation with no greenhouse gas emissions and sustainability because sunlight is abundant and renewable.	The system returned the supported advantages.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-10

localhost:8501

Deploy

What are the advantages of solar energy?

Run Pipeline

Results

✓ FULLY GROUNDED

Grounded ratio: 100% · Answer modified: No — fully grounded · RAGAS PASS

2

GROUNDING CLAIMS

0

HALLUCINATED CLAIMS

100%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation. It is also a sustainable source of energy because sunlight is abundant and renewable. However, solar energy generation depends on weather conditions and is less effective during cloudy days

GROUNDING CLAIMS

- The advantages of solar energy are that it is clean and produces no greenhouse gas emissions during operation
- it is a sustainable source of energy because sunlight is abundant and renewable

FINAL ANSWER (GUARANTEED GROUNDED)

Part 1: The advantages of solar energy are that it is "clean and produces no greenhouse gas emissions during operation" and it is "a sustainable source of energy because sunlight is abundant and renewable".

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-11	Why is solar energy considered clean?	Answer should state that solar energy produces no greenhouse gas emissions during operation.	The system returned the supported clean-energy reason.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-11

localhost:8501

Deploy

Why is solar energy considered clean?

Run Pipeline

 Results

✓ FULLY GROUNDED

Grounded ratio: 100% · Answer modified: No — fully grounded · RAGAS PASS

1

GROUNDING CLAIMS

0

HALLUCINATED CLAIMS

100%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation. It is also a sustainable source of energy because sunlight is abundant and renewable. However, solar energy generation depends on weather conditions and is less effective during cloudy days

GROUNDING CLAIMS

- Solar energy is considered clean because it is clean and produces no greenhouse gas emissions during operation.

FINAL ANSWER (GUARANTEED GROUNDED)

Part 1: Solar energy is considered clean because "it is clean and produces no greenhouse gas emissions during operation."

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-12	What is wind energy?	Answer should remain within the document and describe wind energy as a major renewable energy source generated using wind turbines.	The system kept the supported claim that wind energy is a major renewable source and removed an unsupported claim.	PARTIALLY GROUNDED - hallucinations removed Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-FG-12

localhost:8501

Deploy

What is wind energy?

Run Pipeline

Results

⚠️ PARTIALLY GROUNDED

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore.
Wind energy is efficient and produces no direct

✓ GROUNDED CLAIMS

- Wind energy is another major renewable energy source.

✗ HALLUCINATED CLAIMS (REMOVED)

- This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

Wind energy is another major renewable energy source.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-14	What are wind farms?	Answer should state that wind turbines are often grouped together in wind farms, which can be located on land or offshore.	The system kept the supported wind-farm grouping statement and removed unsupported wording.	PARTIALLY GROUNDED - hallucinations removed Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-FG-14

localhost:8501

Deploy

>>

What are wind farms?

Run Pipeline

Results

PARTIALLY GROUNDED

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore.
Wind energy is efficient and produces no direct

GROUNDED CLAIMS

Wind farms are locations where wind turbines are often grouped together.

HALLUCINATED CLAIMS (REMOVED)

FINAL ANSWER (GUARANTEED GROUNDED)

Wind farms are locations where wind turbines are often grouped together.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-15	Where can wind farms be located?	Answer should state that wind farms can be located on land or offshore.	The system returned the supported location statement.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-15

localhost:8501

Deploy

Where can wind farms be located?

Run Pipeline

 Results

✓ FULLY GROUNDED

Grounded ratio: 100% · Answer modified: No — fully grounded · RAGAS PASS

1

GROUNDED CLAIMS

0

HALLUCINATED CLAIMS

100%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore.
Wind energv is efficient and produces no direct

✓ GROUNDED CLAIMS

- Wind farms can be located on land or offshore.

🔴 FINAL ANSWER (GUARANTEED GROUNDED)

Part 1: Wind farms can be located on land or offshore.

Category 2 - Partially Grounded Test Cases: 2-Part Questions

Questions with one supported part and one unsupported part.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-01	What is solar energy and how efficient are modern solar panels?	Answer the supported solar-energy definition and state that modern panel efficiency is not specified in the context.	The system kept the PV-panel explanation and stated that modern panel efficiency is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00
Screenshot Evidence - TC-PG2-01				

localhost:8501

Deploy

What is solar energy and how efficient are modern solar panels?

Run Pipeline



Results

 **PARTIALLY GROUNDED**

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · **RAGAS PASS**

1

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDED CLAIMS

- Solar energy is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity.

HALLUCINATED CLAIMS (REMOVED)

FINAL ANSWER (GUARANTEED GROUNDED)

Solar energy is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. However, the provided information does not specify the efficiency of modern solar panels.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-02	How is wind energy generated and which country produces the most wind energy?	Answer the supported wind-generation mechanism and state that the country producing the most wind energy is not specified.	The system kept the wind-turbine generation explanation and stated that the country detail is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00
Screenshot Evidence - TC-PG2-02				

localhost:8501

Deploy

How is wind energy generated and which country produces the most wind energy?

Run Pipeline

Results

⚠️ PARTIALLY GROUNDED

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore.
Wind energy is efficient and produces no direct

✓ GROUNDED CLAIMS

Wind energy is generated using wind turbines, which convert the kinetic energy of moving air into electricity.

✗ HALLUCINATED CLAIMS (REMOVED)

🔴 FINAL ANSWER (GUARANTEED GROUNDED)

Wind energy is generated using wind turbines, which convert the kinetic energy of moving air into electricity. However, the provided information does not specify which country produces the most wind energy.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-03	What are the advantages of solar energy and how much does installation cost?	Answer the supported solar-energy advantages and state that installation cost is not specified.	The system returned clean/no-emissions and sustainable sunlight advantages, then stated cost is not specified.	PARTIALLY GROUNDED Grounded Ratio: 67% RAGAS: 1.00

Screenshot Evidence - TC-PG2-03

localhost:8501

Deploy

What are the advantages of solar energy and how much does installation cost?

Run Pipeline



Results

 **PARTIALLY GROUNDED**

Grounded ratio: 67% · Answer modified: Yes — hallucinations removed · RAGAS PASS

2

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

67%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation. It is also a sustainable source of energy because sunlight is abundant and renewable. However, solar energy generation depends on weather conditions and is less effective during cloudy days or at night.

GROUNDED CLAIMS

- The advantages of solar energy are that it is clean and produces no greenhouse gas emissions during operation
- it is a sustainable source of energy because sunlight is abundant and renewable

FINAL ANSWER (GUARANTEED GROUNDED)

The advantages of solar energy are that it is clean and produces no greenhouse gas emissions during operation. Additionally, it is a sustainable source of energy because sunlight is abundant and renewable. However, the cost of installation is not specified.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-04	What are wind farms and how tall is the average wind turbine?	Answer the supported wind-farm description and state that average turbine height is not specified.	The system kept the wind-farm location/grouping information and stated average height is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-PG2-04

localhost:8501

Deploy

What are wind farms and how tall is the average wind turbine?

Run Pipeline

 Results

⚠️ PARTIALLY GROUNDED

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore. Wind energy is efficient and produces no direct

✓ GROUNDED CLAIMS

Wind farms are often grouped together in wind farms, which can be located on land or offshore.

✗ HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context

🗨️ FINAL ANSWER (GUARANTEED GROUNDED)

Wind farms are often grouped together, which can be located on land or offshore. However, the provided information does not specify the average height of a wind turbine.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-05	How do batteries store energy and what is the energy density of lithium-ion batteries?	Answer the supported battery storage role and state that lithium-ion energy density is not specified.	The system kept the energy-storage role and stated that lithium-ion energy density is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-PG2-05

localhost:8501

Deploy

How do batteries store energy and what is the energy density of lithium-ion batteries?

Run Pipeline



Results

 **PARTIALLY GROUNDED**

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Both solar and wind energy play a significant role in reducing reliance on fossil fuels. Governments around the world are investing heavily in renewable energy technologies to combat climate change and reduce carbon emissions. Energy storage systems, such as batteries, are often used alongside renewable energy sources to store

GROUNDED CLAIMS

- Batteries store energy by being used to store excess energy and provide power when production is low.

HALLUCINATED CLAIMS (REMOVED)

FINAL ANSWER (GUARANTEED GROUNDED)

Batteries store energy by being used to store excess energy and provide power when production is low. However, the provided information does not specify the energy density of lithium-ion batteries.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-06	What is solar energy and who invented solar panels?	Answer the supported solar-energy definition and state that the inventor of solar panels is not mentioned.	The system kept the PV-panel explanation and stated that the provided facts do not mention who invented solar panels.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-PG2-06

localhost:8501

Deploy

What is solar energy and who invented solar panels?

Run Pipeline

Results

PARTIALLY GROUNDED

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

1

50%

1.00

GROUNDED CLAIMS

HALLUCINATED CLAIMS

GROUNDED RATIO

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

GROUNDED CLAIMS

HALLUCINATED CLAIMS (REMOVED)

FINAL ANSWER (GUARANTEED GROUNDED)

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-07	How do wind turbines work and what is the global market size of wind energy?	Answer the supported turbine mechanism and state that global wind-energy market size is not mentioned.	The system kept the kinetic-energy conversion explanation and stated that market size is not mentioned.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-PG2-07

localhost:8501

Deploy

How do wind turbines work and what is the global market size of wind energy?

Run Pipeline

Results

PARTIALLY GROUNDED

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

GROUNDING CLAIMS

FINAL ANSWER (GUARANTEED GROUNDED)

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore.

Wind energy is efficient and produces no direct

- Wind turbines work by converting the kinetic energy of moving air into electricity using wind turbines.

Wind turbines work by converting the kinetic energy of moving air into electricity using wind turbines. The provided information does not mention the global market size of wind energy.

HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-08	What makes solar energy clean and what percentage of global electricity comes from solar?	Answer the supported clean-energy reason and state that the global electricity percentage is not specified.	The system kept the no-greenhouse-gas-emissions claim and stated the solar electricity percentage is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-PG2-08

localhost:8501

Deploy

What makes solar energy clean and what percentage of global electricity comes from solar?

Run Pipeline



Results

 **PARTIALLY GROUNDED**

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · **RAGAS PASS**

1

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation. It is also a sustainable source of energy because sunlight is abundant and renewable. However, solar energy generation depends on weather conditions and is less effective during cloudy days or at night.

GROUNDED CLAIMS

- Solar energy is clean because it produces no greenhouse gas emissions during operation.

HALLUCINATED CLAIMS (REMOVED)

FINAL ANSWER (GUARANTEED GROUNDED)

Solar energy is considered clean because it produces no greenhouse gas emissions during operation. However, the provided information does not specify what percentage of global electricity comes from solar.

Category 2 - Partially Grounded Test Cases: 3-Part Questions

Questions with one or two supported parts and other unsupported details.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG3-01	What is solar energy, how efficient are modern panels, and what does installation cost?	Keep the supported solar-energy explanation and reject missing efficiency and installation-cost details.	The system kept the PV-panel explanation and stated that efficiency and cost are not specified.	PARTIALLY GROUNDED Grounded Ratio: 33% RAGAS: 1.00
Screenshot Evidence - TC-PG3-01				

localhost:8501

Deploy

What is solar energy, how efficient are modern panels, and what does installation cost?

Run Pipeline

1

Grounded Claims

2

Hallucinated Claims

33%

Grounded Ratio

1.00

RAGAS Faithfulness

Retrieved Context

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

Grounded Claims

Solar energy is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity.

Hallucinated Claims (Removed)

Final Answer (Guaranteed Grounded)

Solar energy is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. However, the provided information does not specify the efficiency of modern panels or the cost of installation.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG3-02	How do wind turbines work, which country has the most wind farms, and how tall is the tallest turbine?	Keep the supported turbine mechanism and reject country-ranking and tallest-turbine-height details.	The system kept the wind-turbine mechanism and stated that the country and tallest-height details are not specified.	PARTIALLY GROUNDED Grounded Ratio: 33% RAGAS: 1.00

Screenshot Evidence - TC-PG3-02

localhost:8501

Deploy

How do wind turbines work, which country has the most wind farms, and how tall is the tallest turbine?

Run Pipeline

Results

PARTIALLY GROUNDED

Grounded ratio: 33% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDING CLAIMS

2

HALLUCINATED CLAIMS

33%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

GROUNDING CLAIMS

FINAL ANSWER (GUARANTEED GROUNDED)

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore.

Wind energy is efficient and produces no direct

- Wind turbines work by converting the kinetic energy of moving air into electricity using wind turbines.

Wind turbines work by converting the kinetic energy of moving air into electricity using wind turbines. However, the provided information does not specify which country has the most wind farms or the height of the tallest turbine.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG3-03	What are the advantages of solar energy, who invented PV panels, and what is the average payback period?	Keep the supported solar advantages and reject inventor and payback-period details.	The system kept the supported solar advantages and stated that inventor and payback-period information are not provided.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00
Screenshot Evidence - TC-PG3-03				

localhost:8501

Deploy

What are the advantages of solar energy, who invented PV panels, and what is the average payback period?

Run Pipeline

Results

PARTIALLY GROUNDED

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · RAGAS PASS

2

GROUNDING CLAIMS

2

HALLUCINATED CLAIMS

50%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation. It is also a sustainable source of energy because sunlight is abundant and renewable. However, solar energy generation depends on weather conditions and is less effective during cloudy days or at night.

Title: Renewable Energy - Solar and Wind Power

GROUNDING CLAIMS

- The advantages of solar energy are that it is clean and produces no greenhouse gas emissions during operation
- it is a sustainable source of energy because sunlight is abundant and renewable

FINAL ANSWER (GUARANTEED GROUNDED)

The advantages of solar energy include being a clean source of energy, as it produces no greenhouse gas emissions during operation. Additionally, it is a sustainable source of energy because sunlight is abundant and renewable. However, the information provided does not mention who invented PV panels or the average payback period.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG3-04	What is renewable energy, what is nuclear energy, and what is tidal energy?	Keep only renewable-energy information supported by the solar/wind context and reject nuclear and tidal energy details.	The system provided renewable-energy information from the context and stated nuclear/tidal details are not available.	PARTIALLY GROUNDED Grounded Ratio: 33% RAGAS: 0.91

Screenshot Evidence - TC-PG3-04

localhost:8501

Deploy

What is renewable energy, what is nuclear energy, and what is tidal energy?

Run Pipeline

Results

PARTIALLY GROUNDED

Grounded ratio: 33% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDING CLAIMS

2

HALLUCINATED CLAIMS

33%

GROUNDING RATIO

0.91

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDING CLAIMS

- Renewable energy includes Solar and Wind Power as stated in the title, and it is generated from sources such as sunlight and wind.

HALLUCINATED CLAIMS (REMOVED)

FINAL ANSWER (GUARANTEED GROUNDED)

Renewable energy is generated from sources such as sunlight and wind, and it includes Solar and Wind Power. As for nuclear energy and tidal energy, there is no information provided about these topics. Therefore, I can only provide information about renewable energy, which is generated from natural sources like sunlight and wind, including Solar and Wind Power.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG3-05	How is solar energy generated, what is the cost per watt, and who are the top manufacturers?	Keep the supported solar-generation explanation and reject cost-per-watt and manufacturer details.	The system kept the PV-panel generation explanation and stated that cost per watt and top manufacturers are not included.	PARTIALLY GROUNDED Grounded Ratio: 33% RAGAS: 1.00

Screenshot Evidence - TC-PG3-05

localhost:8501

Deploy

How is solar energy generated, what is the cost per watt, and who are the top manufacturers?

Run Pipeline



Results

 **PARTIALLY GROUNDED**

Grounded ratio: 33% · Answer modified: Yes — hallucinations removed · **RAGAS PASS**

1

GROUNDED CLAIMS

2

HALLUCINATED CLAIMS

33%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

 GROUNDED CLAIMS

- Solar energy is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity.

 HALLUCINATED CLAIMS (REMOVED)

FINAL ANSWER (GUARANTEED GROUNDED)

Solar energy is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. However, the provided information does not include details on the cost per watt or the top manufacturers of solar energy systems.

Category 2 - Partially Grounded Test Cases: 4-Part Questions

Questions with one supported part and three unsupported parts.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG4-01	What is solar energy, who invented it, how efficient are modern panels, and what does it cost to install?	Keep the supported solar-energy generation information and reject inventor, efficiency, and installation-cost details.	The system kept the PV-panel generation explanation and stated that the other three details are not specified.	PARTIALLY GROUNDED Grounded Ratio: 25% RAGAS: 1.00

Screenshot Evidence - TC-PG4-01

localhost:8501

Deploy

>>

What is solar energy, who invented it, how efficient are modern panels, and what does it cost to install?

Run Pipeline

Results

PARTIALLY GROUNDED

Grounded ratio: 25% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDED CLAIMS

3

HALLUCINATED CLAIMS

25%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDED CLAIMS

Solar energy is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity.

HALLUCINATED CLAIMS (REMOVED)

FINAL ANSWER (GUARANTEED GROUNDED)

Solar energy is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Unfortunately, the information provided does not mention who invented solar energy. Additionally, the efficiency of modern panels and the cost to install them are not specified in the given facts.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG4-02	How do wind turbines work, how tall is the tallest one, which country leads in wind energy, and what is the global capacity?	Keep the supported wind-turbine mechanism and reject tallest-height, leading-country, and global-capacity details.	The system kept the wind-turbine mechanism and stated that the other three details are not included.	PARTIALLY GROUNDED Grounded Ratio: 25% RAGAS: 1.00

Screenshot Evidence - TC-PG4-02

localhost:8501

Deploy

How do wind turbines work, how tall is the tallest one, which country leads in wind energy, and what is the global capacity?

Run Pipeline



Results

 **PARTIALLY GROUNDED**

Grounded ratio: 25% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDED CLAIMS

3

HALLUCINATED CLAIMS

25%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore.
Wind enerav is efficient and produces no direct

GROUNDED CLAIMS

- Wind turbines work by converting the kinetic energy of moving air into electricity using wind turbines.

HALLUCINATED CLAIMS (REMOVED)

FINAL ANSWER (GUARANTEED GROUNDED)

Wind turbines work by converting the kinetic energy of moving air into electricity using wind turbines. Unfortunately, the provided information does not include details about the height of the tallest wind turbine, the country leading in wind energy, or the global capacity.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG4-03	What is renewable energy, what is nuclear energy, what is tidal energy, and what is geothermal energy?	Keep only the renewable-energy information supported by the solar/wind context and reject nuclear, tidal, and geothermal details.	The system kept renewable-energy information and stated that nuclear, tidal, and geothermal energy are not covered by the context.	PARTIALLY GROUNDED Grounded Ratio: 25% RAGAS: 0.91

Screenshot Evidence - TC-PG4-03

localhost:8501

Deploy

>>

What is renewable energy, what is nuclear energy, what is tidal energy, and what is geothermal energy?

Run Pipeline

Results

PARTIALLY GROUNDED

Grounded ratio: 25% · Answer modified: Yes — hallucinations removed ·

RAGAS PASS

1

GROUNDED CLAIMS

3

HALLUCINATED CLAIMS

25%

GROUNDED RATIO

0.91

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDED CLAIMS

Renewable energy includes Solar and Wind Power as stated in the title, and it is generated from sources such as sunlight and wind.

HALLUCINATED CLAIMS (REMOVED)

FINAL ANSWER (GUARANTEED GROUNDED)

Renewable energy is generated from sources such as sunlight and wind, and it includes Solar and Wind Power. As for nuclear energy, tidal energy, and geothermal energy, the provided context does not offer any information about these topics. Therefore, I can only provide information about renewable energy, which is generated from sources like sunlight and wind, including Solar and Wind

Category 3 - Out of Context Test Cases: Completely Unrelated Topics

Questions fully outside the uploaded solar and wind energy context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-U01	What is nuclear energy?	Reject the question because nuclear energy is outside the uploaded solar/wind context.	The system returned that a reliable answer cannot be provided based on the provided context.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-U01

localhost:8501

Deploy

What is nuclear energy?

Run Pipeline

Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · **RAGAS FAIL**

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

0.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDING CLAIMS

None found.

HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-U02	How does hydroelectric power work?	Reject the question because hydroelectric power is not in the provided context.	The system returned that a reliable answer cannot be provided based on the provided context.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-U02

localhost:8501

Deploy

How does hydroelectric power work?

Run Pipeline

Results

✖ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

0

GROUNDING CLAIMS

1

1

HALLUCINATED CLAIMS

0%

0%

GROUNDING RATIO

0.00

0.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore.
Wind energy is efficient and produces no direct

✓ GROUNDING CLAIMS

None found.

✖ HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

🗨️ FINAL ANSWER (GUARANTEED GROUNDING)

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-U03	What is tidal energy?	Reject the question because tidal energy is outside the solar/wind context.	The system returned that a reliable answer cannot be provided based on the provided context.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.50

Screenshot Evidence - TC-OOC-U03

localhost:8501

Deploy

Run Pipeline

Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

0.50

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

FINAL ANSWER (GUARANTEED GROUNDED)

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

✓ GROUNDED CLAIMS

None found.

✗ HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

I cannot provide a reliable answer based on the provided context.

Category 3 - Out of Context Test Cases: Factual Questions Not in the Document

Real-world factual questions not available in the uploaded document.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F01	Who invented solar panels?	Reject the question because the context does not mention the inventor of solar panels.	The system avoided answering from outside knowledge and returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 1.00

Screenshot Evidence - TC-OOC-F01

localhost:8501

Deploy

Who invented solar panels?

Run Pipeline

Results

OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS PASS

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

FINAL ANSWER (GUARANTEED GROUNDED)

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

None found.

HALLUCINATED CLAIMS (REMOVED)
This information is not in the provided context.

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F02	What is the payback period for a home solar installation?	Reject the question because payback-period information is not provided.	The system returned that a reliable answer cannot be provided based on the provided context.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-F02

localhost:8501

Deploy

What is the payback period for a home solar installation?

Run Pipeline

Results

OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

0

GROUNDING CLAIMS

1

1

HALLUCINATED CLAIMS

0%

0%

GROUNDING RATIO

0.00

0.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation. It is also a sustainable source of energy because sunlight is abundant and renewable. However, solar energy generation depends on weather conditions and is less effective during cloudy days or at night.

GROUNDING CLAIMS

None found.

HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDING)

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F03	How many solar panels does the average home need?	Reject the question because household panel-count requirements are not provided.	The system returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-F03

localhost:8501

Deploy

How many solar panels does the average home need?

Run Pipeline

Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

0.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

FINAL ANSWER (GUARANTEED GROUNDED)

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms

None found.

✗ HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F04	When were solar panels first invented?	Reject the question because historical invention dates are not in the context.	The system returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-F04

localhost:8501

Deploy

When were solar panels first invented?

Run Pipeline

Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

0.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy — Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDING CLAIMS

None found.

HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F05	What is the average efficiency of modern solar panels?	Reject the question because modern solar-panel efficiency values are not provided.	The system returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-F05



>>

Deploy

What is the average efficiency of modern solar panels?

Run Pipeline

Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

0.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

✓ GROUNDING CLAIMS

None found.

✗ HALLUCINATED CLAIMS (REMOVED)

- This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDING)

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F06	What does a solar panel installation cost?	Reject the question because solar installation cost is not specified.	The system refused to provide unsupported pricing details.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 1.00

Screenshot Evidence - TC-OOC-F06

localhost:8501

Deploy

Run Pipeline

Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS PASS

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDING CLAIMS

None found.

HALLUCINATED CLAIMS (REMOVED)

- This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F07	Which country produces the most solar energy in the world?	Reject the question because country-level production rankings are not in the context.	The system returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-F07

localhost:8501

Deploy

>>

Which country produces the most solar energy in the world?

Run Pipeline

Results

✖ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

0.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDING CLAIMS

None found.

HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F08	How tall is the tallest wind turbine in the world?	Reject the question because tallest-turbine height is not provided.	The system returned that a reliable answer cannot be provided from the context.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 1.00

Screenshot Evidence - TC-OOC-F08

localhost:8501

Deploy

>>

How tall is the tallest wind turbine in the world?

Run Pipeline

Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS PASS

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore. Wind energy is efficient and produces no direct emissions, making it environmentally friendly.

GROUNDING CLAIMS

None found.

HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

I cannot provide a reliable answer based on the provided context.

91°F Mostly sunny

Search

2:27 PM 4/29/2026

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F09	Which country has the most wind farms?	Reject the question because country rankings for wind farms are not provided.	The system returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 1.00

Screenshot Evidence - TC-OOC-F09

localhost:8501

Deploy

Which country has the most wind farms?

Run Pipeline

 Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS PASS

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore.
Wind energy is efficient and produces no direct emissions making it environmentally friendly

GROUNDING CLAIMS

None found.

HALLUCINATED CLAIMS (REMOVED)

- This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F10	What is the global market size of the solar energy industry?	Reject the question because market-size data is not provided.	The system returned that a reliable answer cannot be produced from the context.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.50

Screenshot Evidence - TC-OOC-F10

localhost:8501

Deploy

What is the global market size of the solar energy industry?

Run Pipeline

Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

0.50

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDING CLAIMS

None found.

HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F11	What is the energy density of lithium-ion batteries?	Reject the question because lithium-ion battery energy density is not specified.	The system returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 1.00

Screenshot Evidence - TC-OOC-F11

← → ↺ ⓘ localhost:8501 ☆ R ⋮

>> Stop Deploy ⋮

Run Pipeline

Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS PASS

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Both solar and wind energy play a significant role in reducing reliance on fossil fuels. Governments around the world are investing heavily in renewable energy technologies to combat climate change and reduce carbon emissions. Energy storage systems, such as batteries, are often used alongside renewable energy sources to store

GROUNDING CLAIMS

None found.

✗ HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

I cannot provide a reliable answer based on the provided context.

Category 4 - Mixed Multi-Part Stress Test

Prompts combining supported and unsupported question parts.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-MIX-01	What is the cost of solar panels and how are they installed on rooftops?	Keep the supported rooftop/large solar farm installation-location claim and reject cost and detailed installation-process claims.	The system kept the supported rooftop/solar-farm location and stated that cost/process details are not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00
Screenshot Evidence - TC-MIX-01				

localhost:8501

Deploy

What is the cost of solar panels and how are they installed on rooftops?

Run Pipeline

Results

PARTIALLY GROUNDED

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDED CLAIMS

Solar panels are commonly installed on rooftops or in large solar farms.

HALLUCINATED CLAIMS (REMOVED)

FINAL ANSWER (GUARANTEED GROUNDED)

The provided information does not mention the cost of solar panels. However, it does state that solar panels are commonly installed on rooftops or in large solar farms. This indicates that rooftop installation is a common practice for solar panels, but it does not provide details on the installation process or the associated costs.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-MIX-02	Who invented wind turbines and how do they convert kinetic energy into electricity?	Keep the supported kinetic-energy conversion claim and reject inventor details.	The system kept the wind-turbine conversion claim and stated that the inventor is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00
Screenshot Evidence - TC-MIX-02				

localhost:8501

Deploy

Who invented wind turbines and how do they convert kinetic energy into electricity?

Run Pipeline

Results

PARTIALLY GROUNDED

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

GROUNDING CLAIMS

FINAL ANSWER (GUARANTEED GROUNDED)

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore. Wind energy is efficient and produces no direct

- Wind turbines convert the kinetic energy of moving air into electricity.

HALLUCINATED CLAIMS (REMOVED)

Wind turbines convert the kinetic energy of moving air into electricity. However, the information provided does not specify who invented wind turbines.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-MIX-03	What percentage of global electricity comes from solar and what are the advantages of solar energy?	Keep the supported solar-energy advantages and reject global electricity percentage details.	The system kept the supported advantages and stated that the percentage is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-MIX-03

localhost:8501

Deploy

What percentage of global electricity comes from solar and what are the advantages of solar energy?

Run Pipeline

Results

⚠️

PARTIALLY GROUNDED

Grounded ratio: 50%

Answer modified: Yes — hallucinations removed

RAGAS PASS

1

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation. It is also a sustainable source of energy because sunlight is abundant and renewable. However, solar energy generation depends on weather conditions and is less effective during cloudy days or at night.

✓ GROUNDED CLAIMS

The advantages of solar energy are that it is clean and produces no greenhouse gas emissions during operation and it is a sustainable source of energy because sunlight is abundant and renewable.

✗ HALLUCINATED CLAIMS (REMOVED)

FINAL ANSWER (GUARANTEED GROUNDED)

The percentage of global electricity that comes from solar is not specified in the provided information. However, the advantages of solar energy are that it is clean and produces no greenhouse gas emissions during operation, and it is a sustainable source of energy because sunlight is abundant and renewable.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-MIX-04	How efficient are modern solar panels and do they produce greenhouse gas emissions during operation?	Keep the supported no-greenhouse-gas-emissions claim and reject modern panel efficiency details.	The system kept the supported no-emissions claim, but RAGAS faithfulness was lower for this mixed response.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 0.67

Screenshot Evidence - TC-MIX-04

localhost:8501
☆ R ⋮

>>
Deploy ⋮

Results

PARTIALLY GROUNDED

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

1

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDED RATIO

0.67

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

GROUNDED CLAIMS

FINAL ANSWER (GUARANTEED GROUNDED)

One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation. It is also a sustainable source of energy because sunlight is abundant and renewable. However, solar energy generation depends on weather conditions and is less effective during cloudy days or at night.

- It is stated that solar energy produces no greenhouse gas emissions during operation

HALLUCINATED CLAIMS (REMOVED)

Modern solar panels are efficient in the sense that they produce no greenhouse gas emissions during operation. This means that once they are operational, they do not contribute to greenhouse gas emissions.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-MIX-05	What is the tallest wind turbine and can wind farms be located offshore?	Keep the supported offshore/land location claim and reject tallest-turbine details.	The system kept the wind-farm location claim and stated that tallest-turbine information is not mentioned.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-MIX-05

localhost:8501

☆ R New Chrome available

>>

Deploy

What is the tallest wind turbine and can wind farms be located offshore?

Run Pipeline

 Results

⚠ PARTIALLY GROUNDED

Grounded ratio: 50% · Answer modified: Yes — hallucinations removed · RAGAS PASS

1

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

50%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore.
Wind energy is efficient and produces no direct

✓ GROUNDED CLAIMS

• Wind farms can be located on land or offshore.

✗ HALLUCINATED CLAIMS (REMOVED)

• This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

Wind farms can be located on land or offshore. However, the information provided does not mention the tallest wind turbine.

Category 5 - Edge Cases: Trick Questions

Leading or contradiction-based questions that should not be accepted blindly.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-T01	Solar energy is generated at night using stored PV energy, right?	Reject the leading claim because the context does not support solar energy being generated at night using stored PV energy.	The system refused to provide the unsupported leading claim as a grounded answer.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-EDGE-T01

localhost:8501

Deploy

Solar energy is generated at night using stored PV energy, right?

Run Pipeline

 Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

0.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation. It is also a sustainable source of energy because sunlight is abundant and renewable. However, solar energy generation depends on weather conditions and is less effective during cloudy days or at night.

GROUNDING CLAIMS

None found.

HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-T02	Wind turbines only work onshore, correct?	Reject the trick claim because the context says wind farms can be located on land or offshore.	The system removed the unsupported “only onshore” claim and returned a context-limited response.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-EDGE-T02

localhost:8501

Deploy

>>

Wind turbines only work onshore, correct?

Run Pipeline

Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

0.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

GROUNDING CLAIMS

FINAL ANSWER (GUARANTEED GROUNDED)

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore. Wind energy is efficient and produces no direct emissions, making it environmentally friendly.

None found.

✗ HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context that says wind turbines 'only' work onshore, as it states they can be located 'on

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-T03	Solar energy produces some greenhouse gas emissions during operation, doesn't it?	Reject the contradiction because the context says solar energy produces no greenhouse gas emissions during operation.	The system did not accept the false assumption and returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-EDGE-T03

localhost:8501

Deploy

Solar energy produces some greenhouse gas emissions during operation, doesn't it?

Run Pipeline

 Results

✖ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

0.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation. It is also a sustainable source of energy because sunlight is abundant and renewable. However, solar energy generation depends on weather conditions and is less effective during cloudy days or at night.

GROUNDING CLAIMS

None found.

✖ HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context that supports the statement. Instead it says "it is clean and produces no

FINAL ANSWER (GUARANTEED GROUNDED)

I cannot provide a reliable answer based on the provided context.

Category 5 - Edge Cases: Ambiguous Questions

Broad questions that can lead to over-answering.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-A01	Tell me everything about renewable energy.	Answer only with renewable-energy facts present in the context and avoid expanding beyond the document.	The system returned a partially grounded answer with supported solar/wind facts and removed unsupported expansion.	PARTIALLY GROUNDED Grounded Ratio: 67% RAGAS: 1.00

Screenshot Evidence - TC-EDGE-A01

localhost:8501

Deploy

Tell me everything about renewable energy.

Run Pipeline

Results

PARTIALLY GROUNDED

Grounded ratio: 67%

Answer modified: Yes — hallucinations removed

RAGAS PASS

2

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

67%

GROUNDED RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDED CLAIMS

Solar energy is one of the most widely used renewable energy sources in the world, generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity.

Both solar and wind energy play a significant role in reducing reliance on fossil fuels, with governments investing heavily in

FINAL ANSWER (GUARANTEED GROUNDED)

Renewable energy is an important aspect of reducing our reliance on fossil fuels. One of the most widely used renewable energy sources in the world is solar energy. This type of energy is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. In addition to solar energy, wind energy also plays a significant role in this effort.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-A02	What are all the advantages and disadvantages of both solar and wind energy?	Summarize only advantages and disadvantages explicitly found in the context.	The system used most supported solar/wind advantage and disadvantage claims; one unsupported claim was removed.	PARTIALLY GROUNDED Grounded Ratio: 83% RAGAS: 0.86

Screenshot Evidence - TC-EDGE-A02

localhost:8501

Deploy

What are all the advantages and disadvantages of both solar and wind energy?

Run Pipeline

Results

PARTIALLY GROUNDED

Grounded ratio: 83% · Answer modified: Yes — hallucinations removed · RAGAS PASS

5

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

83%

GROUNDED RATIO

0.86

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation. It is also a sustainable source of energy because sunlight is abundant and renewable. However, solar energy generation depends on weather conditions and is less effective during cloudy days or at night.
Title: Renewable Energy – Solar and Wind Power

GROUNDED CLAIMS

- One of the main advantages of solar energy is that it is clean and produces no greenhouse gas emissions during operation.
- It is also a sustainable source of energy because sunlight is abundant and renewable.
- Solar energy generation depends on weather conditions and is less effective during cloudy days or at night.

FINAL ANSWER (GUARANTEED GROUNDED)

The advantages of solar energy include being clean and producing no greenhouse gas emissions during operation, as well as being a sustainable source of energy due to the abundance and renewability of sunlight. On the other hand, one of the disadvantages of solar energy is that its generation depends on weather conditions, making it less effective during cloudy days or at night.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-A03	Compare solar and wind energy across all dimensions.	Compare only across dimensions found in the context, such as generation, installation/location, environmental impact, and storage support.	The system produced a mostly supported comparison, but unsupported content was identified and removed.	PARTIALLY GROUNDED Grounded Ratio: 83% RAGAS: 0.96

Screenshot Evidence - TC-EDGE-A03

localhost:8501

New Chrome available

>>

Deploy

Compare solar and wind energy across all dimensions.

Run Pipeline

 Results

 PARTIALLY GROUNDED

Grounded ratio: 83% · Answer modified: Yes — hallucinations removed · RAGAS PASS

5

GROUNDED CLAIMS

1

HALLUCINATED CLAIMS

83%

GROUNDED RATIO

0.96

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on

✓ GROUNDED CLAIMS

- Solar energy is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity, whereas wind energy is generated using wind turbines, which convert the kinetic energy of moving air into electricity.

🔍 FINAL ANSWER (GUARANTEED GROUNDED)

Solar and wind energy can be compared across various dimensions. In terms of generation, solar energy is produced by capturing sunlight using photovoltaic panels, which convert sunlight directly into electricity. On the other hand, wind energy is generated using wind turbines, which convert the kinetic energy of moving air into electricity.

Category 5 - Edge Cases: Questions That Sound Like They Are in the Document But Are Not

Related-sounding questions that request missing factual details.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-S01	How many solar panels are installed worldwide?	Reject the question because the context does not include a worldwide solar-panel count.	The system returned a context-limited refusal instead of inventing a number.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-EDGE-S01

localhost:8501

New Chrome available

>>

Deploy

How many solar panels are installed worldwide?

Run Pipeline

 Results

✗ OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

0.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDING CLAIMS

None found.

✗ HALLUCINATED CLAIMS (REMOVED)

- This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-S02	What percentage of homes use solar energy?	Reject the question because household solar adoption statistics are not provided.	The system refused to provide an unsupported percentage.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-EDGE-S02

localhost:8501

Deploy

What percentage of homes use solar energy?

Run Pipeline

Results

OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS FAIL

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

0.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Title: Renewable Energy – Solar and Wind Power
Solar energy is one of the most widely used renewable energy sources in the world. It is generated by capturing sunlight using photovoltaic (PV) panels, which convert sunlight directly into electricity. Solar panels are commonly installed on rooftops or in large solar farms.

GROUNDING CLAIMS

None found.

HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDING)

I cannot provide a reliable answer based on the provided context.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-S03	How much noise do wind turbines produce in decibels?	Reject the question because the context only states that wind turbines can be noisy; it does not provide decibel values.	The system refused to provide an unsupported decibel value.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 1.00

Screenshot Evidence - TC-EDGE-S03

localhost:8501

Deploy

How much noise do wind turbines produce in decibels?

Run Pipeline

Results

OUT OF CONTEXT

Grounded ratio: 0% · Answer modified: Yes — hallucinations removed · RAGAS PASS

0

GROUNDING CLAIMS

1

HALLUCINATED CLAIMS

0%

GROUNDING RATIO

1.00

RAGAS FAITHFULNESS

RETRIEVED CONTEXT

Wind energy is another major renewable energy source. It is generated using wind turbines, which convert the kinetic energy of moving air into electricity. Wind turbines are often grouped together in wind farms, which can be located on land or offshore.
Wind energy is efficient and produces no direct emissions, making it environmentally friendly.

GROUNDING CLAIMS

None found.

HALLUCINATED CLAIMS (REMOVED)

This information is not in the provided context.

FINAL ANSWER (GUARANTEED GROUNDED)

I cannot provide a reliable answer based on the provided context.

